

When a quantum link keeps changing, who decides how to run it?

A satellite quantum link never holds still — the satellite moves, the air flickers, clouds pass. This poster is about whether a system can learn to run such a link well while the ground keeps shifting under it, and what a first honest test says.

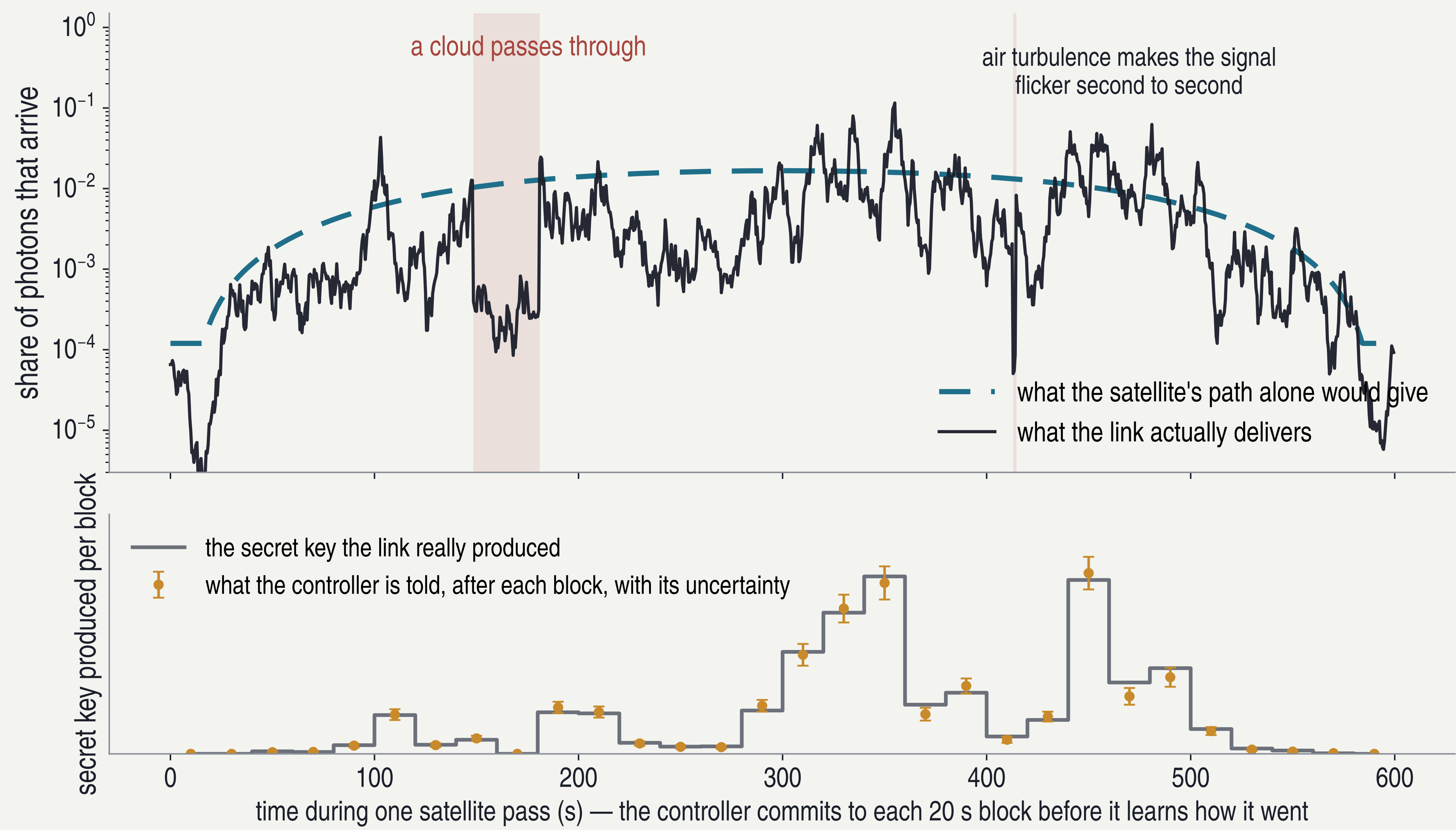
Programme title: Quantum-QKD-Aero — An Open Research Platform for Adaptive Machine Learning in Quantum Communication

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One pass: the environment keeps changing

ILLUSTRATIVE



Above: the share of photons that make it from satellite to ground during one ten-minute pass. The dashed curve is what the satellite's path alone would give; the real link flickers with the air and drops out when a cloud crosses. Below: the link runs in blocks, and the controller finds out how each block went only afterwards, and only approximately.

The controller has to know which changes matter

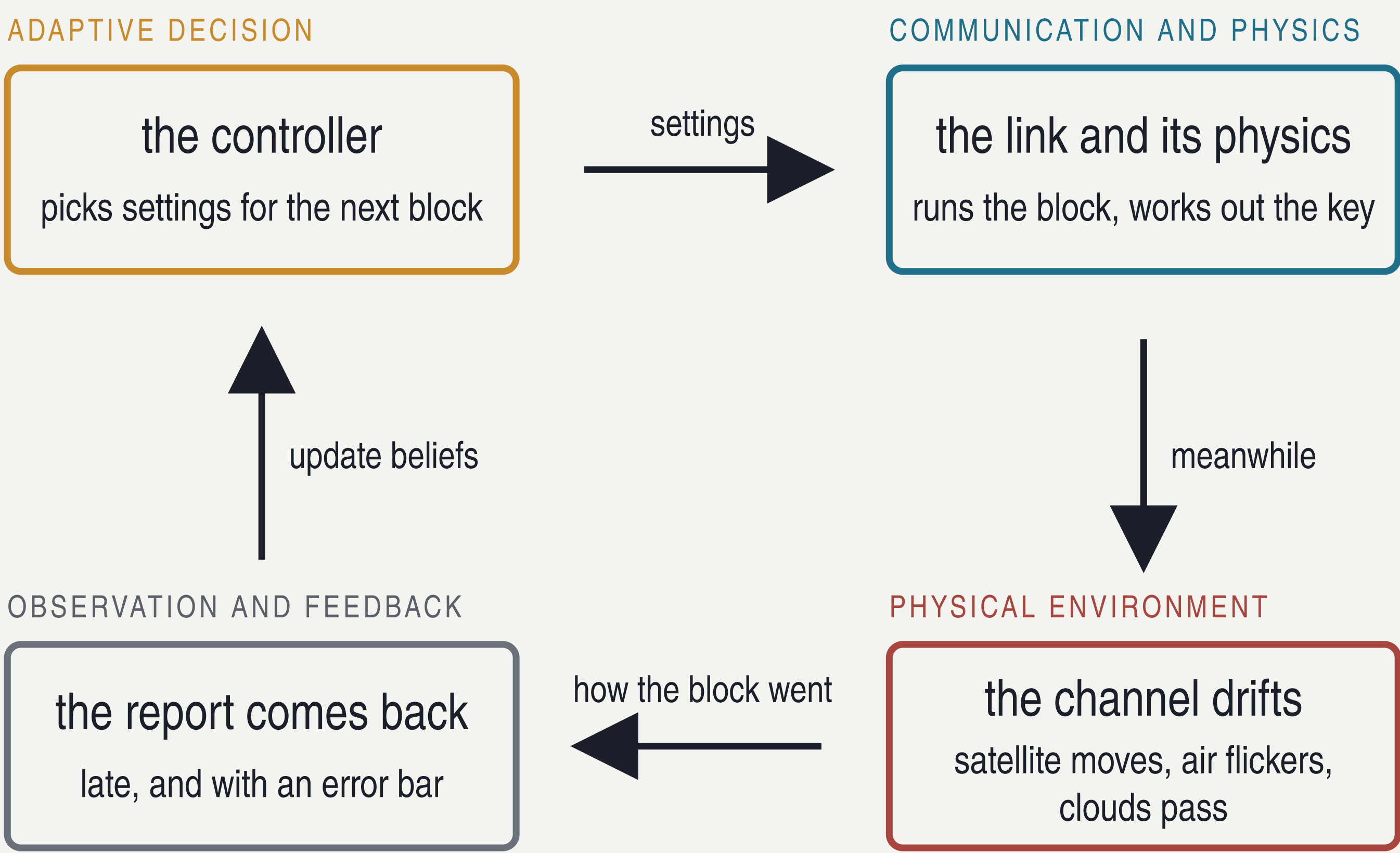
- Conditions move while the controller is still learning.
- Feedback arrives only after the controller has committed to a block.
- Short blocks give faster but noisier answers; long blocks give clearer answers about conditions that may already have changed.

Temporary fluctuation
The controller should hold steady.
Reacting costs performance.

Meaningful change
The controller should adapt.
Waiting costs performance.

The central problem is deciding which of the two the system is looking at. Anyone keeping a quantum device on its operating point while it drifts faces the same question; a simulated link is a good place to study it because the physical assumptions are explicit and checkable, and holding them fixed makes controller failures easier to diagnose.

One controller: the loop it lives in



Each cycle the controller picks settings, the link runs, the physics reports back how much key was made and how sure it is, and the controller updates what it believes — while the channel has already moved on.

What the controller can see and do

It sees	How high the satellite is, and how the previous block went.
It chooses	How bright to make the transmitter, and how long to run before pausing to take stock.
It is scored on	The secret key the link certifies for each block, reported after the block ends, with an error bar.
It is compared with	A fixed setting that never changes during a pass; a reactive controller that adjusts to the last block but remembers nothing; and a look-ahead controller allowed to peek at the next block's conditions.

Why the separation matters

BUILT

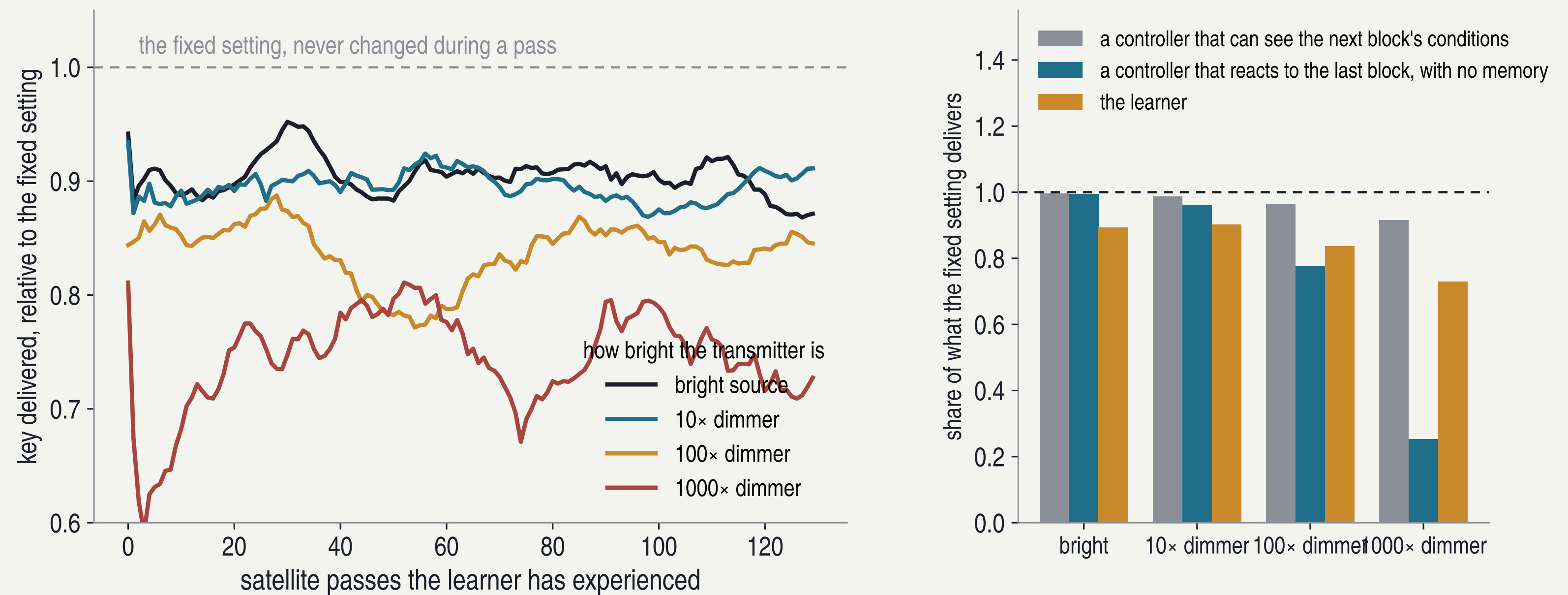
Quantum-QKD-Aero keeps the physical link separate from the communication logic and from anything that learns. Every controller in the study is scored against exactly the same open-source simulator. Its physical models are tested against known answers, and each output carries provenance showing where it came from. The same separation lets later studies add nodes, routes, memories, and network-level controllers without letting adaptation rewrite the physics it is being tested against.

There is a wall between the physics and anything that learns. The learner can ask the physics questions; it cannot reach in and change the answers.

First result: was adapting worth it?

SIMULATED, PRELIMINARY

With transmitter brightness and block length as its only adaptive decisions, the learner did not beat the fixed setting.



A simple learner was given one hundred passes to practise and thirty to be scored on, and the whole test was repeated with the transmitter turned down to make the link harder. Everything is measured against the fixed setting. Run in September 2026 on the open simulator, with the learning code kept outside the physics.

- The reactive controller chased noise, and did dramatically worse as the link got dimmer.
- The learner was more stable, but it never overcame the cost of frequent adjustment.
- The next study adds decisions a fixed setting cannot make: sitting out a cloud, and responding to interference.

BUILT physical simulator and baselines RUNNING adaptive-link study NEXT sit-out and interference decisions

Many networks: from one changing link to quantum internetworking

Quantum-QKD-Aero begins with one changing link. The longer-term aim is to study how multiple quantum nodes, and eventually different kinds of quantum networks, can coordinate when information is delayed, resources are limited, and no participant can see the whole system. Delayed, noisy, and incomplete feedback does not go away at network scale; it becomes a coordination problem.

